

Chikku Voice AI Kiosk

1. Product Overview

Chikku Voice AI Kiosk is a smart humanoid kiosk robot solution that functions as a professional virtual assistant. The standout feature of Chikku is its ability to engage in natural, real-time voice interactions, providing customers with a seamless information retrieval experience. Primary deployment target environments include: **Restaurants, Hotels, Supermarkets, and Shopping Malls.**

2. Core Architecture & Technologies

The Chikku system operates on a **Hybrid RAG** architecture, combining two advanced retrieval methods to process complex queries:

- **Vector Search (Semantic Retrieval):** Handles the diversity in user phrasing, enabling the robot to accurately understand standard Q&A intent (e.g., dish descriptions, policy inquiries).
- **Graph Traversal (Relational Retrieval):** Maps connections between entities using a **Knowledge Graph**, allowing the robot to reason and precisely answer spatial or cross-logical questions (e.g., *"Which menu items contain shrimp?"* or *"Which store closest to me sells children's toys?"*).

3. Features & Use Cases

Chikku is designed to be customized for specific deployment environments:

For Restaurants & F&B Services

- Detailed Q&A regarding menus, ingredients, flavors, and allergy information.
- Provides up-to-date pricing structures and restaurant policies.
- Analyzes and recommends dishes based on preferred ingredients or tastes.

For Supermarkets & Shopping Malls

- Provides floor plans and spatial positioning information.
- Offers turn-by-turn navigation to specific zones or retail booths.
- Enables location queries by product category (identifying the nearest booth based on the user's current location).

4. Operations & Integration

- **Dynamic Data Synchronization:** The system integrates directly with Point of Sale (POS) software. This allows Chikku to automatically update rapidly changing data in real time, such as dish/product stock levels and time-sensitive pricing strategies (e.g., Happy Hour).

S-Robot

1. Product Overview

- **Definition:** S-Robot is a specialized robot for cleaning solar panels.
- **Objective:** Automate the maintenance process to keep the surface of solar panels consistently clean.
- **Value Proposition:** Maximize solar energy absorption efficiency and power output.
- **Deployment Scale:** Rooftop solar installations (residential, commercial) and utility-scale solar farms.

2. Core Application

- **Primary Mission:** Replace human labor in cleaning dust, sand, fallen leaves, and bird droppings from panel glass surfaces.
- **Occupational Safety:** Eliminate risks associated with manual cleaning in hazardous terrains, high rooftops, and steep slopes.
- **Operational Assurance:** Maintain continuous, stable efficiency across solar arrays.

3. Core Values & Benefits

Time Savings & Efficiency

- Operates automatically according to set schedules without requiring human supervision.
- Cleans significantly faster than manual methods, enabling power plants to restore peak output immediately.

Cost Savings

- Cuts costs related to specialized cleaning labor and personal protective equipment (PPE).
- Prevents risks of micro-cracks and glass scratching caused by human foot traffic during manual cleaning.
- Optimizes long-term Return on Investment (ROI) and extends overall system lifespan.

Eco-Friendly

- Optimizes water consumption or utilizes dry-cleaning technology.
- 100% chemical-free operation, protecting anti-reflective coatings and the surrounding ecosystem.

WatchGuard6S

1. Vision & Mission

- **Positioning:** Revolutionize security and maintenance operations in solar farms through cutting-edge technology.
- **Objective:** Ensure sustainable energy production while achieving the highest standards of safety, efficiency, and accuracy.

2. Overview & Core Solution

- **Definition:** WatchGuard6S is an AI-integrated robotic system dedicated to security, surveillance, and real-time monitoring.
- **24/7 Security Protection:** Equipped with high-resolution cameras, thermal imaging sensors, and AI algorithms for continuous patrolling and surveillance.
- **Optimization & Maintenance:** Automatically detects environmental hazards, automates maintenance workflows, and optimizes solar panel operational performance.

3. Technology & Innovativeness

- **Advanced Camera Systems:** Provides sharp visual clarity under all lighting and weather conditions, including thermal imaging capabilities.
- **AI-Driven Analytics:** Processes images and data instantaneously to detect anomalies, security intrusions, or equipment failures.
- **Multifunctional Sensors:** Comprehensively collects environmental metrics, temperature data, and the physical status of the power station.
- **Autonomous Operation:** Capable of self-patrolling, autonomous navigation, and executing missions independently without human intervention.